

Blending Synthetic Anomalies for Anomaly Detection in Human X-ray imaging

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Abstract— X-rays have proven to be a crucial technology in the context of security applications, due to its ability to inspect contents of objects through non-invasive means. The downside of this technology, however, is that it requires manual monitoring and is susceptible to human error. In this study we explore methods of anomaly detection that may aid in the inspection of X-ray imaging. Our approach focuses on exploring the effects of using synthetic anomalies when training the teacher module of the Student-Teacher architecture. We therefore also introduce in this study a novel method of blending synthetic anomalies into X-ray images such that they appear natural where they are introduced. Our study finds that the use of synthetic anomalies for the teacher module of the S-T architecture is effective in detecting anomalies, but how hard to detect the synthetic anomalies are has almost no effect. However, we find that our method of introducing synthetic anomalies is versatile enough to imitate real anomalies in certain cases.

Index Terms—Anomaly Detection, Synthetic Anomaly, X-ray Imaging, Security X-rays

1 INTRODUCTION

The ability of X-ray technology to inspect the contents carried by people and objects without physical contact or intrusion has made it a standout choice among other methods of content analysis [1]. In particular, the application of X-rays for this purpose has made it indispensable for security applications in facilities such as airports, government facilities, and correctional facilities. With this technology, these facilities can not only identify potential security threats, but also inspect areas otherwise challenging to examine without invasive or, in the case of objects like baggage and containers, destructive means.

Given the urgency and precision necessary in determining threats, it becomes apparent that quickly analysing items in X-ray scans is crucial. Additional factors like human subjectivity and fatigue during analysis also make the need for automatic and reliable image processing evident. This has led to the exploration of recent advances in the fields of Computer Vision (CV), Machine Learning, and Deep Learning, which have certainly brought forward the possibility of creating such systems.

For instance, particularly in medical imaging, researchers have devised techniques that allow them to create supportive systems to help medical professionals detect anomalies in medical X-ray scans [2]. Various methods have also been developed to successfully apply CV as a security measure for detecting and identifying threats that go through baggage scanners typically used in airports and similar facilities [3]. Still, challenges remain in the automation of X-ray security analysis.

In the context of using X-rays to generate body scans as a security measure, the human body introduces complexities due to the variability in composition and size of individuals. These scans need to be analysed quickly, but often without the controlled setting of a medical room to standardise the data, and without a predefined target to identify. Certain regions of these scans may also be problematic

during analysis. Specifically, the abdominal region, which can have a wide variation of density and contents, presents unique challenges that highlight the need for sophisticated analysis techniques. We therefore explore in our study how we can address the issue of variability in this data through Anomaly Detection methods. In particular, we explore how pretraining on different representations of anomalies can affect anomaly detection performance.

Before this can be achieved, the problem of X-ray data abundance in the security context must also be considered. Because individuals carrying security threats is not a common occurrence, the distribution of X-ray data in this context is naturally skewed towards normal X-ray scans. Therefore, we include in our discussion a method of creating synthetic anomalies to mimic the diverse range of items that people may carry when going through X-ray scans.

In light of these challenges, our primary objective is to leverage state-of-the-art CV techniques in developing a system that assists operators in identifying foreign items, especially in denser regions of full-body X-ray scans. We also aim to fill in the crucial gap of missing anomalous data in security X-ray imaging. Through the following research questions, we achieve this goal:

- How does using synthetic anomalies in training the teacher model of a Student-Teacher architecture affect the performance of anomaly detection models on real anomalies?
- How can we create synthetic anomalies in the chest and stomach areas of security X-ray imaging, such that they blend naturally into regions in which they are introduced?
- How can we comprehensively consider a range of different anomalies that may be encountered in the security context of X-ray imaging in our methodology?

We begin our approach by overviews the current state of CV based on the challenges previously discussed (Section 2). This is followed by a detailed description of the datasets we utilised coupled with the innovative technologies and methodologies that underpinned our experiments (Section 3). We then showcase the results of our experiments, emphasising their significance to our primary research objectives (Section 4). Finally, this study concludes in Section 5, where we delve into the broader consequences of our discoveries and discuss the future trajectory of our work.

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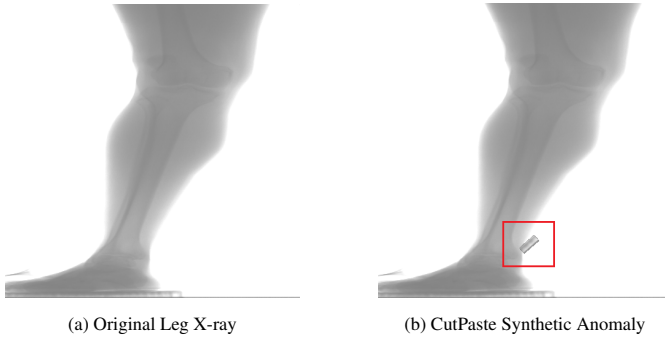


Fig. 1: CutPaste [4] method of introducing synthetic anomalies to normal data. The anomaly in (1b) is added to the normal image where ankle monitors are commonly found in anomalous images. The anomaly is not smoothed into the patch it is introduced, making it stand out further.

2 STATE OF THE ART

We explore in this section recent developments in the field of Computer vision. Particularly, the state of anomaly detection and anomaly detection in medical imaging is relevant to our work. This section also discusses current methods of creating synthetic anomalies, and how these methods relate to our work.

2.1 Synthetic Anomaly Generation

Creating synthetic anomalies is often utilised where real data cannot be acquired, or where data annotation is not feasible. For example, in the case of medical anomalies, annotating is often costly and requires experts to perform the annotations. Another case is in the context of security X-rays, where X-ray scans of individuals carrying contraband is not a common occurrence, resulting in scarce data.

Various methods have become prominent in the process of creating synthetic anomalies, among which, patch-based anomaly creation has proven to be an effective way of synthesising anomalies. [4] introduces a straight-forward way of doing so by copying a patch from an image, and pasting it elsewhere in the image as an anomaly. However, the disadvantages of this method are that it results in anomalies that stand out too much, and do not represent actual anomalies that can be encountered in a given domain. [5, 6] perform similar patch-based transformations, but interpolate the anomaly appearance based on the pixel values between the copied patch and the destination patch. This method shares similar disadvantages to the method of [4], where anomalies may stand out if the location they are pasted to is not optimal.

For example, the CutPaste method [4], which relies on the autoencoder-based calculations of common positions of anomalies found in anomalous images, has the possibility of taking a patch of the human body from a full-body X-ray, and applying it outside the human body as an anomaly. In our dataset of full-body human X-rays, while an ankle monitor is commonly found in security X-ray images, the methods of [6, 5, 4] can paste a normal part of the image to the location of an ankle monitor, leaving floating pixels unrelated to the rest of the X-ray. This is demonstrated in Figure 1.

This issue is accounted for in the method described by [7]. Rather than relying on autoencoder-based calculations for common anomaly positions, they instead calculate the anomaly location in to ensure it is in an appropriate position. Figure 2 illustrates the process used by [7] in selecting synthetic anomaly destinations.

While this approach works well for industry-type anomalies, it fails to consider the gradient where the normal image meets the edges of the synthetically added anomaly. An example of this scenario is present in the example given in Figure 1, where the edges of the synthetic anomaly stand out significantly from the location it was added to. [8] address this in their study, where they smooth the edges

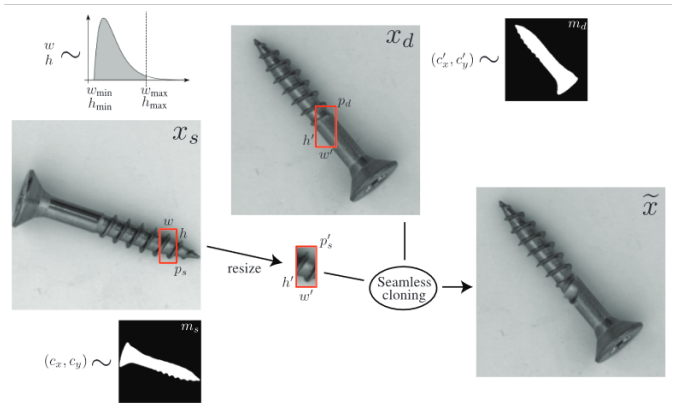


Fig. 2: Process used by [7] to select the destination patch of the synthetic anomaly. First a patch is selected from the source image, and is then resized and pasted to the destination image where the conditions (2) and (3) are satisfied.

in areas where the introduced anomaly meets the rest of the image. By doing this, they stop reliance on the high gradient between synthetic anomalies and the rest of the image. In our study, we employ methodology similar to [8] and [7] that addresses the discussed disadvantages in our context.

2.2 Anomaly Detection Methods

The task of anomaly detection in computer vision has seen a rapid growth in research, owed to its broad applications and the ability to apply it without the need for vast amounts of labelled data. For modern anomaly detection, one of the main methods employed is utilising feature extractors [9] to define normal and anomalous image features. For instance, [10] utilises the WideResNet-101 [11] feature extractor, and [12, 13] utilise output layers of EfficientNet-B5 [14] to extract features. In addition to feature extractors, [15, 16] suggest the use of autoencoders for learning the logical constraints of data. The application of autoencoders extends past the use of industrial inspections indicated by [15], and even into anomaly detection in X-ray imaging, where [17, 18] have developed novel architectures that leverage autoencoders to determine the locations of anomalies in medical images. By using these two methods, the current state of the art (SOTA) is able to detect structural anomalies such as damaged surfaces, and logical anomalies such as unexpected items, missing items, or items of irregular shape and size.

[19] utilises these techniques in their study to detect these types of anomalies with a student-teacher (S-T) architecture. Rather than using an existing feature extractor in their approach, they instead distill a WideResNet-101 [11] feature extractor into a smaller module referred to as a Patch Description Network (PDN). This module consists of only four convolutional layers and is illustrated in Figure 3. Because this module is fully convolutional, a single forward pass can calculate all the output feature vectors of an image. Additionally, the PDN includes a strided average-pooling layers after the first and second convolutional layers, as [20] demonstrated that this reduced the runtime and memory usage of the PDN.

An illustration of the Efficient AD architecture is displayed in Figure 4. It detects structural anomalies by comparing the outputs of the student and the teacher, while relying on the difference between the student and the autoencoder to detect logical anomalies.

Another method that significantly improved SOTA performance was Reverse Distillation [21]. Similarly to [19] they employ the S-T architecture, but rather than directly implementing an autoencoder module with a student and teacher module, they instead modify the S-T architecture to behave similar to an autoencoder. In the traditionally employed S-T architecture, both modules have the same network structure and data flows through either module in the same

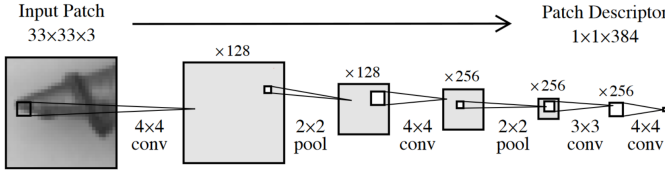


Fig. 3: Patch Description Network (PDN) architecture from Efficient AD [19]. The fully convolutional design of the architecture allows extraction of image features in a single forward pass.

way during distillation. [21] identify these as issues that hinder the generation of discrepant representations between the student and the teacher. Instead, they propose to reverse the shapes of the teacher as an encoder, to create a student as a decoder. Figure 5 illustrates the difference between the traditional S-T architecture and the proposed architecture of [21].

This approach addresses the issue of structural similarity between the student and teacher module. However, the high capacity of teachers in knowledge distillation means that extracted features by high-dimensional descriptors are likely to have considerable redundancy, which can be harmful to the student module in decoding the essential normal features. Additionally, the final encoder block of the teacher is likely to characterise high-level features of the input data. Because of this, feeding the output of the teacher encoder directly into the student can hinder the reconstruction of lower-level features. To address these issues, [21] further employs the use of a one-class bottleneck embedding (OCBE) module, whose architecture is illustrated in Figure 6.

The OCBE module consists of two blocks. The first block is a one-class embedding block (OCE) that projects the teacher module’s high-dimensional representations into a low-dimensional space. This allows the OCE to retain essential information favourable for the student decoder in decoding information from the teacher encoder. The second block is a multiscale feature-fusion block (MMF) that aims to address the problem of low-level feature restoration by concatenating feature representations from different layers of the teacher module. To do this, the MMF block takes features from different layers of the teacher module and applies down sampling convolutions to low-level features to align the representation of low and high-level features.

3 MATERIALS AND METHODS

One key challenge of this research is creating synthetic anomalous data, while ensuring this data is representative of anomalies in the actual dataset. In addressing this challenge, a method of creating artificial anomalies in grayscale images is introduced, and we describe the process of blending them to a natural appearance in section 3.1. We then discuss the dataset utilised in this study and how our data is transformed and divided in section 3.2. The module used, and the reason why this module was selected is discussed in 3.3. Finally, the metrics utilised in assessing our experiments are described in section 3.4.

3.1 Creating Synthetic Anomalies

Various methods of creating synthetic anomalies for the application of industrial anomaly detection are available [4, 7]. One key element utilised by available synthetic anomaly creation methods, however, is that they often rely on existing pixels of the image to create the synthetic anomalies. In the context of anomaly detection in security X-ray scans, where the possibilities of what may be considered anomalies are far more varied, this method can result in anomalies that do not naturally appear as if they belong in the original image. To remedy this and create synthetic anomalies that appear more natural, we propose a new method of blending anomalies into our images. In addition to providing synthetic anomalies with a natural appearance, our methodology has the added benefit of retaining relative information of the background pixels it is added to.

Rather than copying a section of the images and placing it elsewhere, we instead gather and create a collection of images that we classify as anomalies. Examples of these images are shown in Figure 7. The anomaly image A is first converted to greyscale and resized to a new width A_w and height A_h relative to the size of the destination image. The values of A_w and A_h are based on random values selected from a uniform distribution determined by width I_w and height I_h of their destination images divided by pre-selected divisors Min_w , Max_w , Min_h , and Max_h :

$$A_w \sim U(I_w/\text{Min}_w, I_w/\text{Max}_w), \quad A_h \sim U(I_h/\text{Min}_h, I_h/\text{Max}_h).$$

The anomaly image A is then randomly rotated according to an angle $\angle \sim U(0, 360)$ to further increase the variability of introduced anomalies.

We employ similar methodology to [7] in calculating the positions of anomalies. In their methodology, they calculate the anomaly location using conditional random position selection as follows: Given a source image x_s and a destination image x_d of the same width W and height H , a patch p_s of width w and height h is extracted from x_s . The centre coordinates (c_x, c_y) of the destination patch p_d when p_s is applied to x_d , are then derived where

$$c_x \sim U\left(W\frac{w}{2}, W - W\frac{w}{2}\right), \quad c_y \sim U\left(H\frac{h}{2}, H - H\frac{h}{2}\right) \quad (1)$$

To prevent anomalies from floating in the background when calculating the coordinates of p_d , (1) is applied repeatedly until the object masks m_d of x_d and the object masks $(m_{p_s}$ and $m_{p_d})$ of p_s and p_d meet the conditions

$$(p_d \cap m_d) / (w \times h) > t_{object} \quad (2)$$

$$(m_{p_d} \cap m_{p_s}) / |m_{p_s}| > t_{overlap} \quad (3)$$

where t_{object} represents the threshold of how much overlap is desired between the selected destination patch and the object pixels of the destination image, and $t_{overlap}$ represents the threshold of how much overlap is desired between the object in the source patch and the object in the destination patch. The condition of (2), verbosely, requires that the destination patch should be applied to a part of the destination image containing the object to which a synthetic anomaly is added. Similarly, (3) requires that the objects in the source patch and destination patch should overlap, as the condition of (2) alone is not sufficient in preventing floating anomalies.

In our methodology, the patch into which the anomaly is placed is determined by exploring the image in finding an optimal position. Because we guarantee that the centre of the image contains the region of interest through a keypoint detection module, a patch with the dimensions of the resized anomaly (A_w, A_h) is selected at the centre of the image $(p_{x_{now}} = A_w/2, p_{y_{now}} = A_h/2)$. The average value of pixels in this patch is calculated as $p_{mean_{now}}$ and a die is rolled to select the direction to move the patch in the x-axis $x_{direction} \in \{-1, 0, 1\}$ and the y-axis $y_{direction} \in \{-1, 0, 1\}$, where a value of 0 indicates no movement in that axis. This direction is used to calculate the new position

$$p_{x_{next}} = x_{direction} \times n, \quad p_{y_{next}} = y_{direction} \times n, \quad (4)$$

where n is a constant stride distance to move the patch. For each movement, the new mean value of the patch $p_{mean_{next}}$ is compared to the previous value $p_{mean_{now}}$. We select the coordinates $p_{x_{next}}$ and $p_{y_{next}}$ as the new coordinates of our patch if the following condition is met:

$$|p_{mean_{now}} - p_{mean_{next}}| < t_{movement} \quad (5)$$

Where $t_{movement}$ is the threshold of how much change in the mean pixel value is allowed. If the condition (5) is not met, (4) is applied again with a new, random, $x_{direction}$ and $y_{direction}$. A dice is rolled against thresholds t_{walk} and $t_{direction}$ to determine whether the patch

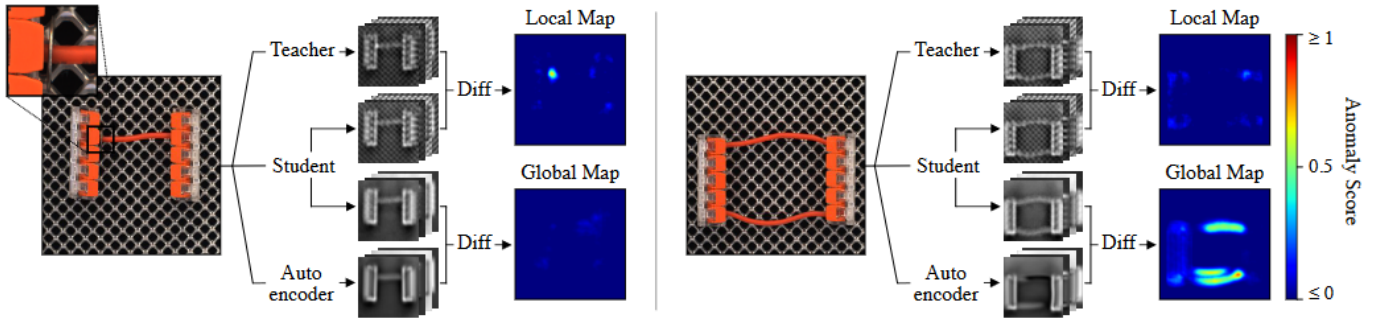


Fig. 4: Architecture of Efficient AD. It incorporates the Student-Teacher architecture to detect structural anomalies, while introducing an autoencoder module to detect logical anomalies. It detects structural anomalies by comparing the output feature maps of the student and the teacher (left), and logical anomalies by comparing extra output channels of the student with the autoencoder reconstruction of the image (right).

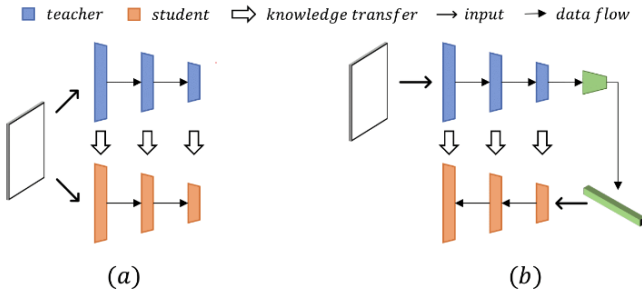


Fig. 5: Comparison between traditional knowledge distillation for S-T architectures (a), and proposed reverse distillation method proposed by [21] (b).

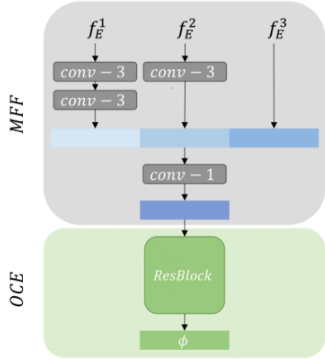


Fig. 6: One Class Bottleneck Embedding module used by [21]. It consists of an MMF and OCE block, where the MMF block aligns multiscale features from teacher E , and the OCE block condenses the obtained rich features to a compact bottleneck ϕ . The output of this module is then passed to the student decoder module.



Fig. 7: Examples of images blended into normal images as anomalies.

should stop moving and whether the direction should change. The conditions are calculated as follows:

$$R \times (n_{walk}/100) < t_{walk}$$

$$D \times (n_{direction}/100) < t_{direction}$$

where R and D are random variables from the set $\{0, 100\}$, n_{walk} is the number of times the patch has been moved and $n_{direction}$ is the number of times the patch has moved since the direction was changed. We set $n_{direction}$ to 0 each time the direction is changed to ensure the patch is moved an adequate distance before another direction change occurs. Setting $t_{direction}$ higher places the patch further from the starting position at the centre, while setting it lower means the patch will stay closer to its starting position. Setting t_{walk} higher means the patch will search for a position for more iterations.

By selecting the anomaly position this way, we explore different possible positions for an anomaly while ensuring the anomaly is located within the object of interest. This method also ensures the anomaly is placed in a location with evenly distributed pixel values and does not cross high-gradient changes, allowing for more natural synthetic anomalies.

Once the final position of the anomaly is determined, the mean intensity of the patch p and the anomaly image A is calculated as p_{mean} and A_{mean} , together with the minimum and maximum values of p , p_{min} and p_{max} . The values of the anomaly image A are then min-max normalised to the range of p_{min} and p_{max} to ensure it has the same distribution as the patch. The pixel values a_{xy} of A are transformed

$$a'_{xy} = p_{mean} + (a_{xy} - A_{mean}) \times (a_{xy\alpha}/Max_{\alpha}) \times (1 - d) \quad (6)$$

where a'_{xy} is the transformed intensity value of each pixel at position (x, y) of A , A_{mean} is the mean intensity value of all the pixels in A , and Max_{α} is the maximum possible value of the alpha channel of A . The alpha values $a_{xy\alpha}$ of each pixel from the anomaly image are then used to further transform each pixel. We finally apply a difficulty factor d as a control that modulates how seamlessly the anomaly is blended into the destination patch. Figure 7 illustrates how the added anomaly differs between incremental difficulty factors. To apply the anomaly to the patch, the values of each derived anomaly pixel a'_{xy} are subtracted from the corresponding pixel of the patch p_{xy} .

We demonstrate the effectiveness of this method in Figure 8 by comparing a key in an unmodified anomalous image, to a key synthetically added as an anomaly by our methodology.

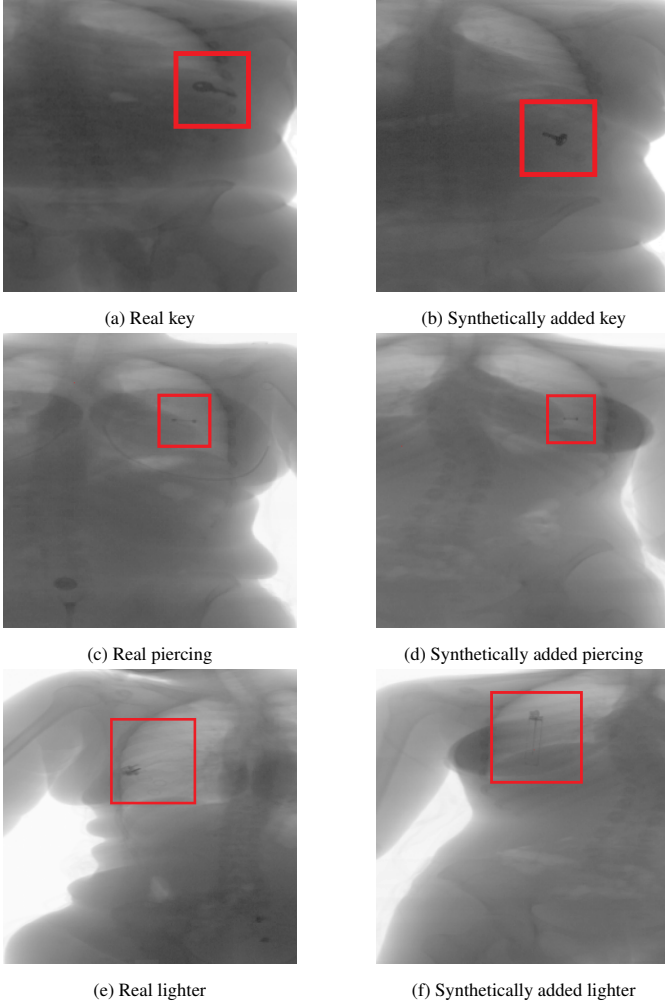


Fig. 8: Comparison between real and synthetic anomalies generated by our method. This demonstrates the ability of our method to imitate specific anomalies. The anomalies displayed in the images are a key (8a, 8b), a piercing (8c, 8d), and a lighter (8e, 8f).

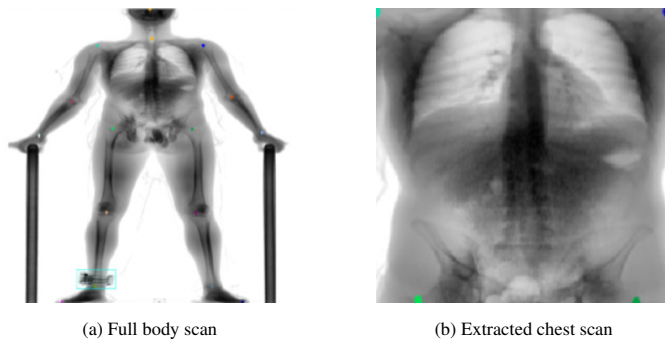


Fig. 9: Chest region extracted from full body X-ray scan. The image on the right (9b) is a subsection of the image on the left (9a), based on the key points detected by our keypoint detection module.

Dataset	clean	anomaly
Training	7,382	0
Pretraining	1,057	1,059
Validation	260	266
Testing _{synthetic}	260	266
Testing _{real}	103	87

Table 1: Distribution of data after synthetic anomaly generation. The training dataset only contains clean images (non-anomalous) to train the modules on what is normal. The testing datasets are divided into synthetic anomalies and real anomalies to test the ability of our synthetic anomalies in mimicking real anomalies.

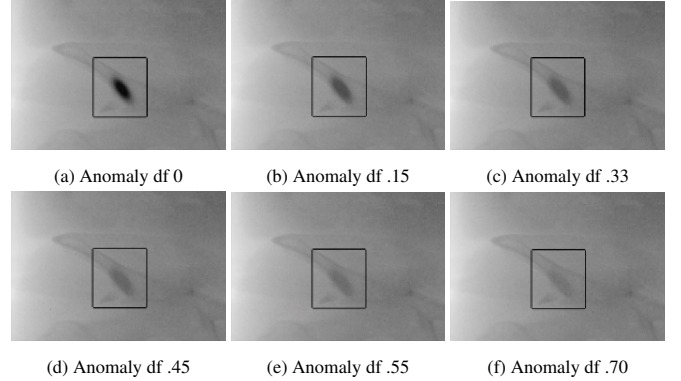


Fig. 10: Synthetic anomalies generated by incremental difficulty factors (df) 0, 0.15, 0.33, 0.45, 0.55, and 0.70.

3.2 Dataset

This study utilises a proprietary dataset containing full-body X-ray scans comprised of 11,131 anomalous images and 3,465 normal images. However, as we wish to focus on the abdominal and chest area of these scans, we employ a keypoint detection module to extract this region. An example of this extraction is demonstrated in Figure 9. Extracting this region allows us to work with the area of our interest without needing special methods of preprocessing to take this into account.

Because the original dataset is not annotated, it is impossible to determine the positions of anomalies. This is important because an anomalous image in the original dataset may not be anomalous in the modified dataset comprised by the extracted chest sections; anomalies may be located in areas other than our area of interest. Therefore, the normal images are the images most suitable for our use. To increase the amount of data available for training, a further 7,096 images among the positive images were selected from the anomaly classes that were deemed to either have an unlikely chance of containing anomalies in our region of interest, or anomalies that may be within our region or interest, but are considered unimportant for the purposes of anomaly detection. An example of this is provided in Figure 9, where an ankle monitor present in Figure 9a is not present in Figure 9b.

The final distribution of our datasets is specified in Table 1. In addition to distributing the dataset as described, we generate several datasets of incremental difficulty factors as described in 3.1. The difficulty factors used in our datasets were 0 as a baseline, and 0.15, 0.33, 0.55 and 0.70 as our incrementally challenging datasets. Across each difficulty factor, each dataset is identical, with the differing factor being how well-blended anomalies are into the original image. An example of a synthetic anomalies generated with different difficulty factors is illustrated in Figure 10.

Because we also want to have a representation of actual anomalous data in our testing metrics, we manually annotate 103 anomalous images of the original dataset for a secondary test set. These images were hand-picked to ensure the anomalies they contained were located in the region of interest for this study.

3.3 Models

While other methods may show potentially better results for Anomaly Detection, such as the method described by, [12], and [21], we also want to test the effects of pretraining with synthetic anomalies on performance against real anomalies. Therefore, we select [19] as our module for testing because the method they describe inherently relies on pretraining a teacher module by distilling knowledge from a WideResNet-101 [11] backbone. By distilling the feature extraction capabilities of this backbone into our module, we can measure the effect of pretraining on synthetic data.

For our implementation of Efficient AD, we evaluated the EfficientAD-S and EfficientAD-M variants per experiment. In pretraining the weights of a WideResNet-101 pretrained on the ImageNet [22] dataset were distilled into a teacher PDN [19]. To adapt the teacher PDN to features specific for our data distribution, we pretrain our teacher on a portion of our dataset generated for this task instead of training the teacher on features of the ImageNet dataset as suggested by [19]. Pretraining was done for 50 epochs, while training was done for 21 epochs with validation done every 3 epochs. For each training process, images were resized to a 448-pixel height and width for consistency in image size. Both normal and anomalous data were utilised for validation, as this allowed calculation of interim anomaly detection and localisation performance.

In our experiments, we further employed the use of PatchCore [10]. This method requires storing all normal images in memory when performing inference, and does not perform validation as normal training methods would. Because of this, instead of utilising the normal training data from our synthetically created datasets, we utilise normal images from the validation dataset for this method. We further reduced the image size transformation to a 370-pixel height and width to address the memory issue. This allowed us to test on both synthetic anomalies and on real anomalies, whilst overcoming the hardware limitation of storing all normal images in-memory to perform inference. We employ the ANOMALIB [23] implementation of PatchCore, for our experiments.

3.4 Performance Metrics

To measure the effectiveness of our synthetic anomalies in teaching modules to distinguish anomalies, we measured performance on a set of real anomalies. The anomaly detection performance was measured with the Area Under the Receiver Operating Characteristic Curve (AU-ROC) by its image-level prediction scores. Localisation performance was measured by and Area Under the Per Region Overlap Curve (AU-PRO) at a false positive rate of 30%, as suggested by [24]. For metrics acquired using the PatchCore method, we instead employ F1-Score at the pixel level to measure localisation performance due to limitations in the AU-PRO implementation of ANOMALIB [23] preventing use of this metric. Calculating either Area Under the Curve requires calculating the True Positive Rate (TPR) in the case of AU-ROC, and the False Positive Rate (FPR) in the case of both AU-ROC and AU-PRO. These metrics can be computed as

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN},$$

where True Positives (TP) are correctly identified anomalies, False Positives (FP) are normal values incorrectly classified as anomalies, True Negatives (TN) are correctly identified normal values, and False Negatives (FN) are anomalies incorrectly classified as normal.

In addition to these, we also consider the F1-Score, which is a harmonic mean of precision and recall. The F1-Score is particularly useful in scenarios where an imbalance between the classes exists. It is calculated as:

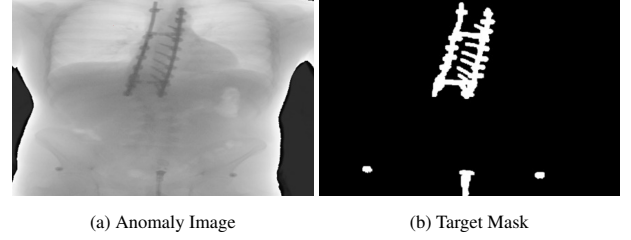


Fig. 11: Cherry-picked image of a real anomaly for a comparison between the different difficulty factors and Efficient AD models. The results of either model are displayed in Figure 12

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where Precision is the ratio of TP to all positive predictions (TP + FP) and Recall is the same as TPR.

3.4.1 Area Under ROC Curve

AU-ROC is a widely used metric for assessing the discriminative ability of a binary classification model, which is particularly relevant in the context of anomaly detection. It is calculated by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification threshold t :

$$AU\text{-ROC} = \int_0^1 TPR(t) d(FPR(t)).$$

3.4.2 Area Under PRO Curve

AU-PRO is a metric specifically designed for comparing the segmentation between a prediction mask and a target mask on the per-region level. For any N number of ground truth components in a dataset, The PRO of any test image in this dataset is computed:

$$PRO = \frac{1}{N} \sum_i \sum_k \frac{|P_i \cap C_{i,k}|}{|C_{i,k}|},$$

where $C_{i,k}$ denotes the set of pixels marked as anomalous for a connected component k in the ground truth image i , and P_i denotes the set of pixels predicted as anomalous at any threshold t .

AU-PRO then is computed by plotting the PRO against the FPR at any threshold t :

$$AU\text{-PRO} = \int_0^1 PRO(t) d(FPR(t)).$$

4 EXPERIMENTS & RESULTS

We develop our experimental design with the goal of assessing the potential of utilising synthetic anomalies to detect various kinds of anomalies in X-ray scans for the security context. In doing these experiments, we aim to test how pretraining anomaly detection methods on synthetic datasets affects performance on real data. We further compare the performance of established models on our synthetic anomalies and real anomalies to assess how well our synthetic anomalies imitate real anomalies.

As discussed above, prior to our experiments, we generated several datasets of synthetic anomalies with incremental difficulty factors. We utilise these datasets in conducting pre-training, training, and testing fully for the selected model.

Tables 2 and 3 display the results of our tests where we pretrained a teacher module on the Efficient AD [19] architecture on various levels of synthetic anomalies. The experiments show that pretraining on different difficulties has little effect on the quantitative results of our metrics. For testing on synthetic anomalies in Table 2, the results showed an AU-ROC at the image level (AU-ROC_{image}) of 0.77 for EfficientAD-S, and 0.82 for EfficientAD-M, on the baseline difficulty

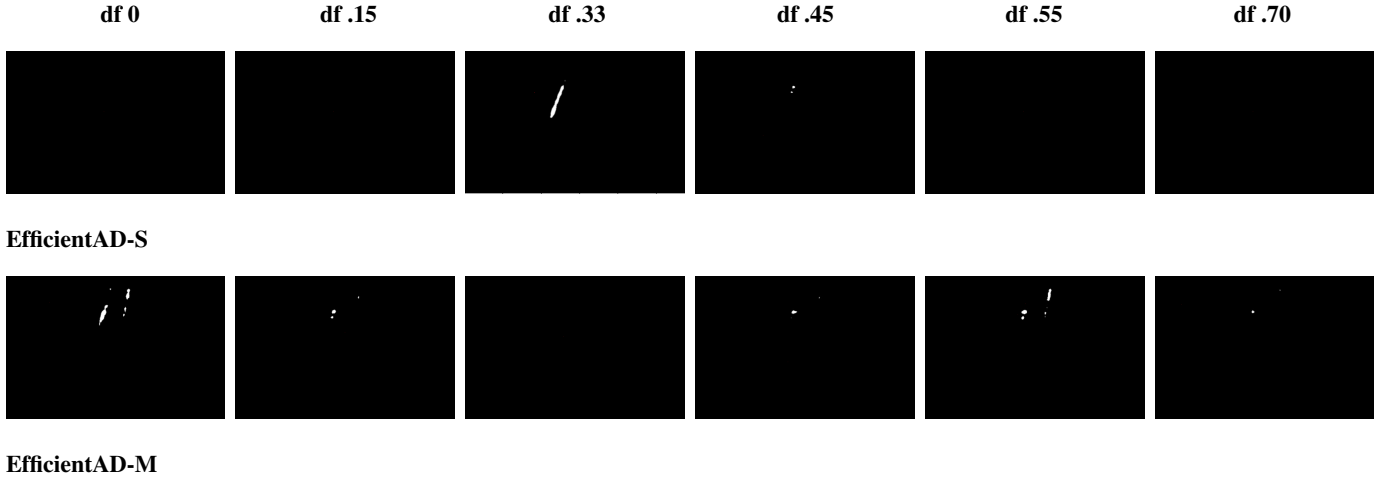


Fig. 12: Prediction masks of Efficient AD models with different pretraining difficulty factors (df). The top row illustrates the prediction masks from the EfficientAD-S model across the different difficulty factors used, and the bottom row illustrates those of the EfficientAD-M models. Each column represents the results after pretraining on each incremental difficulty factor.

Dataset	AU-PRO	AU-ROC _{image}	AU-ROC _{pixel}
EfficientAD-S			
df .0	0.23	0.77	0.55
df .15	0.15	0.55	0.50
df .33	0.23	0.51	0.55
df .45	0.02	0.50	0.50
df .55	0.05	0.55	0.50
df .70	0.05	0.53	0.50
EfficientAD-M			
df .0	0.30	0.82	0.62
df .15	0.03	0.55	0.50
df .33	0.03	0.53	0.50
df .45	0.04	0.52	0.50
df .55	0.04	0.52	0.50
df .70	0.04	0.51	0.50

Table 2: EfficientAD results on synthetic data when pretraining is done on datasets with incremental difficulty factors (df). For either model, performance is best at a difficulty factor of zero, where anomalies are not made less opaque in the image. However, for each incremental difficulty factor, performance quickly drops to random.

factor of 0. Subsequent incremental difficulty showed the performance of either model quickly deteriorates to random. However, comparing the results of a single image against each difficulty factor in pretraining shows that the ability of the model to detect anomalies is affected by the anomalous data it encounters when pretraining. This is demonstrated in Figure 12. This figure illustrates the resulting masks generated at a fixed threshold against a single image containing real anomalies. A fixed threshold was selected for these masks rather than a dynamic threshold for each model. Modifying this threshold showed little difference in the produced mask. This is done because we want to measure how utilising synthetic anomalies of different intensity values can affect the confidence of the model in detecting real anomalies of varied intensity levels as well. Because every step taken for each difficulty factor is identical, it is reasonable to attribute this difference to the synthetic anomalies seen during pretraining. Despite this, Figure 12 demonstrates that there is no significant advantage to utilising the different synthetic datasets in pretraining. We provide the maps of these masks in Appendix B.

Because of the sudden drop in performance observed between difficulty factors 0 and 0.15, we further conducted experiments that

Dataset	AU-PRO	AU-ROC _{image}	AU-ROC _{pixel}
EfficientAD-S			
df .0	0.16	0.53	0.50
df .15	0.15	0.55	0.50
df .33	0.23	0.51	0.51
df .45	0.15	0.65	0.50
df .55	0.15	0.57	0.50
df .70	0.15	0.61	0.50
EfficientAD-M			
df .0	0.17	0.63	0.50
df .15	0.16	0.58	0.50
df .33	0.13	0.53	0.50
df .45	0.16	0.59	0.50
df .55	0.16	0.57	0.50
df .70	0.17	0.58	0.50

Table 3: EfficientAD results on real data when pretraining is done on datasets with incremental difficulty factors (df). Unlike Figure 2, results vary per difficulty factor. Notable differences are the high values of EfficientAD-S for df 0.45, and EfficientAD-M for df 0. However, this difference is not significant and unlikely to be attributed to the difficulty factors used

explored the trend of how this performance decline occurs. Datasets of difficulty factors 0.01, 0.03, 0.05, 0.07, 0.10 and 0.13 were created to observe this trend, and EfficientAD-S was used to test how this performance declines. These difficulty factors were established through observing the random performance of EfficientAD-S at the difficulty factor of 0.15, indicating potentially better performance with lower difficulty factors. Illustrations of how anomalies of these difficulty factors appear can be found in Appendix A. Table 4 provides the results of this experiment, where the gradual decline of model performance is observed as the difficulty factor applied to synthetic anomalies is increased. The anomaly detection performance on Efficient AD on synthetic data reaches that of random at the difficulty factor of 0.10. Its anomaly detection performance on real data, however, remains similar regardless of which synthetic dataset was used when training the teacher module. While this demonstrates the lack of connection between how difficult anomalies are to detect and the capability of the model to detect real anomalies, it does show the capabilities of our method to imitate anomalies. This is particularly true for anomalies which are harder to detect, which is a large concern in the context of security X-rays. Thus, it further allows

us to test the capabilities of other models to detect hard-to-detect anomalies.

Dataset	AU-PRO _s	AU-ROC _s	AU-PRO _r	AU-RUC _r
df .01	0.19	0.79	0.15	0.60
df .03	0.17	0.75	0.15	0.61
df .05	0.15	0.68	0.15	0.57
df .07	0.14	0.59	0.16	0.56
df .10	0.05	0.52	0.16	0.57
df .13	0.05	0.52	0.15	0.57

Table 4: Results of EfficientAD-S on synthetic anomalies (left, s) and real anomalies (right, r) when the teacher module is trained on synthetic datasets with incremental difficulty factors (df).

We use the method described in PatchCore [10] to test this, as it detects anomalies through different means. Unlike the previous method used, which inherently relied on anomalous data for training the teacher module, [10] does not utilise anomalous data at all in achieving inference. We can therefore test this method on our data, as well as on real data, to determine the capabilities of this method to detect obscure anomalies. The results of this experiment are described in Table 5. As this demonstrates, PatchCore outperforms Efficient AD in identifying both synthetic and real anomalies. We observe from these results that the decline in performance is more gradual, and that performance declines to random at the difficulty factor of 0.70. A similarity in performance between real and synthetic anomalies is present at the difficulty factor between .15 and .33 using PatchCore; while for Efficient AD this was observed between difficulty factors 0.07 and 0.10.

Dataset	F1-Score _{pixel}	AU-ROC _{image}
df .0	0.22	0.79
df .01	0.24	0.81
df .03	0.24	0.81
df .05	0.23	0.80
df .07	0.22	0.80
df .10	0.22	0.79
df .13	0.21	0.78
df .15	0.19	0.76
df .33	0.13	0.69
df .45	0.09	0.63
df .55	0.04	0.57
df .70	0.01	0.51
real	0.27	0.74

Table 5: Results of PatchCore on generated synthetic anomalies of incremental difficulty factors (df) and real anomalies (real).

5 DISCUSSION, CONCLUSION & FUTURE WORK

We conclude our study by discussing in Section 5.1 the main contributions of our study, and why these contributions are significant to the overall field. The main conclusions of our study are then drawn in relation to our research questions in Section 5.2. We finally close our study by discussing the limitations of our work and what future research can be done to improve upon it in Section 5.3.

5.1 Discussion

We observe in our experiments that our method of blending synthetic anomalies does have the capability of allowing anomaly detection methods to detect real anomalies. However, the extent to which it is capable of representing real anomalies needs to be further explored. This is due to the lack of any clear correlation between how utilising different difficulty factors of anomalies affects the performance of the model on real anomalies.

We suspect that this method of introducing synthetic anomalies may be capable of imitating real anomalies, given the nature of the

algorithm. Because it does not rely on existing parts of the image, it becomes more versatile in the kinds of anomalies it may introduce in images. The consequence of this would be that it allows, particularly in the context of security, calibrating models to detect anomalies that are obscure. For example, this method of creating synthetic anomalies may enable imitating anomalies that are scarce in the real world, such as drug bags and other harmful substances.

Despite this, we were able to identify two main limitations to our method: Because of how it is implemented, it relies on the user to acquire or generate images that are suitable for blending into the images. For complex objects or objects that have several variations, such as ankle monitors or even lighters (as shown in Figure 8f), this may take significant amounts of time to perform. However, for anomalies that do not attempt to imitate specific objects, such as the one illustrated in 10, this method is successful in addressing the gaps of other methods, as previously discussed. The other limitation of our method is that it may be insufficient in creating enough variations in the introduced anomalies to prevent over-fitting to the specific anomalies.

Another limitation was that testing on the effects of pretraining was done solely with Efficient AD. It is possible that the effects of pretraining utilising different difficulty factors may affect how the model detects anomalies, but Efficient AD was not able to capture these intricacies. Additionally, the feature extractor used for pretraining the PDNs was one trained on the ImageNet dataset. Other feature extractors specialised for the type of dataset we utilise may prove to better represent the features of X-ray imaging than the WideResNet-101 feature extractor.

The final limitation of our study was that synthetic anomalies were generated solely at a single difficulty factor per dataset. While this may help measure the ability of a model to detect anomalies at that given difficulty factor, it is not holistic in representing the different anomalies that may be encountered in a deployed setting.

5.2 Conclusion

Our experiments discovered that focusing on how difficult anomalies are to detect when training the teacher in the Student-Teacher architecture has almost no effect on the ability of the model to detect real anomalies. While variations in the difficulty factor did affect model performance, the nature of this effect is currently unclear in this study.

Despite this, we have developed a method of creating synthetic anomalies that has the capability of imitating and potentially representing real anomalies in the context of security X-ray imaging. The method of introducing anomalies described in this study has shown that it is capable of imitating real anomalies and giving synthetic anomalies an appearance that matches the pixel values of the environment into which they are introduced. It further successfully considers the types of anomalies that may be encountered in both the range of items that may be considered anomalies, and the range of how challenging anomalies may be to detect in X-ray imaging.

5.3 Future Work

Based on the limitations discussed above, future work built upon our discoveries may consider taking these limitations into account for further experiments. For example, the proposed method has shown to be effective in imitating real anomalies, opening the possibility for supervised or semi-supervised methods of detecting anomalies.

This method of introducing synthetic anomalies may also be further improved by incorporating generative methods in creating the anomalies introduced into the images. As generative models [25] continue to improve, creating images that represent anomalies has certainly become a possibility. Though the capability of these models in generating anomalous images directly is still limited by the niche nature of this data, it can still be incorporated into our methodology. In our study, the images used as anomalies were limited by the availability of these images, and the work necessary to create these images by hand. By incorporating generative models to create these

anomaly images, it is possible to improve not only the types of anomalies that may be introduced, but also their variance.

Another work that may be explored is foundation models. As foundation models continue to improve their capabilities of generalising, it may also be worth exploring this for anomaly detection. [26, 27, 28] introduce novel methods of integrating few-shot foundation models for anomaly detection, which may have the capacity to detect anomalies in the context of X-ray security scans. While [29] show that few-shot methods are not as performant as methods that are fully trained on a dataset, the ability of the model to generalise may be sufficient in detecting anomalies in X-ray images.

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A EXTRA SYNTHETIC ANOMALY DIFFICULTY FACTORS

Synthetic anomalies of difficulty factors .01, .03, .05, .07, .10, and .13, which were used to explore the decline of EfficientAD-S performance on synthetic anomalies.

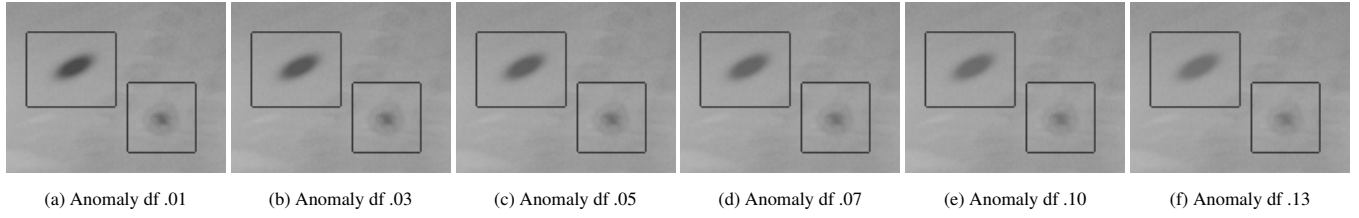


Fig. 13: Synthetic anomalies generated by incremental difficulty factors (df) 0.01, 0.03, 0.05, 0.07, 0.10, and 0.13.

B PREDICTION MAPS

Corresponding prediction maps of the masks shown in Figure 12.

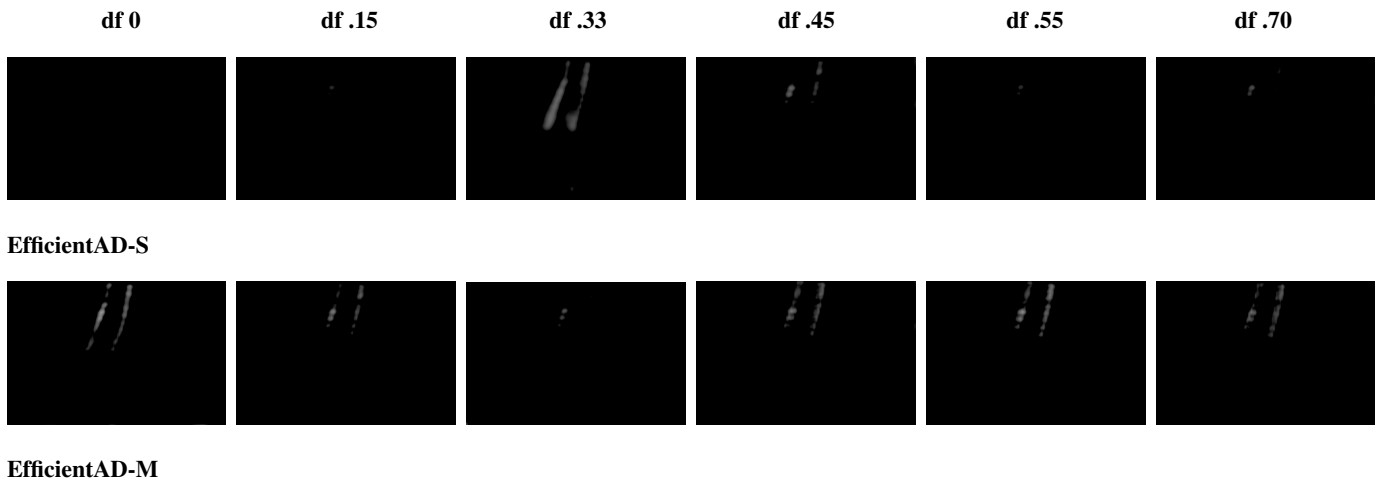


Fig. 14: Prediction maps of Efficient AD models with different pretraining difficulty factors (df). The top row illustrates the prediction masks from the EfficientAD-S model across the different difficulty factors used, and the bottom row illustrates those of the EfficientAD-M models. Each column represents the results after pretraining on each incremental difficulty factor.